

An overview of force measurement: sensing methods, multi-axis measurement, dynamic and static measurement, force estimation and applications.

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Abstract—The abstract goes here.

Index Terms—Interaction Force, Force Measurement.

I. Introduction

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II. Force Sensing Methodes

There are many methods to measure forces exerted on objects. These methods differ from one another in the physical principle of sensing. In this section, we will explain some of these methods and compare between advantages and disadvantages of each method. As shown in Table (??).

A. Resistive force transducer

It is also called the resistive strain gauge. It is an instrument that converts mechanical force into a small change in electrical resistance as shown in figure(1). This

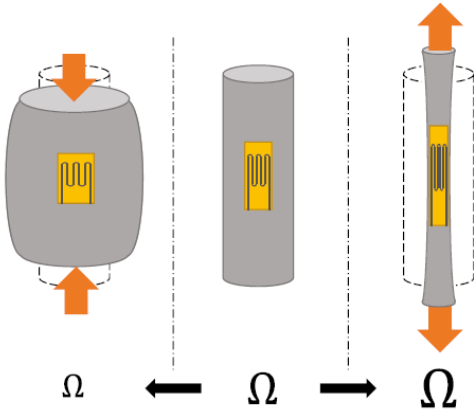


Fig. 1: resistive force transducer [1]

tool consists of a metal foil glued to a thin insulating layer. The metal foil is arranged in a zigzag pattern. The gauge is bonded to the most deformed area of the body subjected to the force. A fundamental parameter of the strain gauge is its gauge factor and nominal gauge resistance. Gauge factor is the sensitivity to strain, it is defined as (1).

$$G_F = \frac{\frac{\Delta R}{R}}{\frac{\Delta L}{L}} = \frac{\frac{\Delta R}{R}}{\varepsilon} \quad (1)$$

where R, L and ε is the original resistance, length and strain of the strain gauge respectively. $\Delta R, \Delta L$ are the resistance and the length variation of the strain gauge because of its deformation. Nominal resistance is the resistance of a strain gauge in an unstrained position. It has various values such as $120 \Omega, 350 \Omega$ and 1000Ω . To convert small variation of resistance into an electrical voltage, Wheatstone bridge circuit is used. The full-bridge circuit consists of four strain gauges. It doubles sensitivity and performs thermal compensation. The sensitivity of full-bridge circuit is defined in (2):

$$\frac{V_o}{V_E} = \varepsilon G_F \quad (2)$$

where V_o is the output voltage of the circuit and V_E is the voltage of the excitation source. In the literature, Al-Dahiree et al.[2] designed the resistive strain gauge load cell. It was designed for measuring the axial force of laboratory test benches. The advanced load cell provides a 300 N load capacity, low-cost material, low weight, and size with highly accurate force measurement. Hensh et al.[3] designed and developed a self-propelled rice seeder for wet fields. With the developed machine, both the required torque and the corresponding power requirements increased as the forward speed increased. Yaun et al.[4] designed and developed a multi-degree-of-freedom force and torque sensor. This sensor was used for installation at the base of a humanoid robot's foot. The developed sensor is capable of measuring vertical forces up to 1000 N and horizontal forces up to 400 N.

- B. Capacitive force transducer
- C. Inductive force transducer
- D. Piezoelectric force transducer
- E. Fiber optic force transducer
- F. Vibrating wire force transducer

III. Static vs Dynamic Force Measurement

IV. Multi-Dimensional Force Measurement

V. Force Estimation

VI. Force Measurement Applications

- A. Force measurement in monitoring and testing tasks
- B. Use of force measurement in automated systems
- C. Force measurement in research and development laboratories

VII. Conclusion

The conclusion goes here.

References

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